basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ELECTRICAL TECHNOLOGY: DIGITAL ELECTRONICS

MAY/JUNE 2026

MARKS: 200

TIME: 3 hours

This question paper consists of 22 pages, a 1-page formula sheet and a 7-page answer sheet.

Copyright reserved

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions.

2. Answer ALL the questions.

3. Answer the following questions on the attached ANSWER SHEETS:

   QUESTIONS 3.3.3, 3.5.2, 3.6.2 and 3.8.2
   QUESTIONS 5.4.1, 5.4.2, 5.5.2, 5.5.3, 5.7 and 5.10
   QUESTIONS 6.8

4. Write your centre number and examination number on every ANSWER SHEET and hand them in with your ANSWER BOOK, whether you have used them or not.

5. Sketches and diagrams must be large, neat and FULLY LABELLED.

6. Show ALL calculations and round off answers correctly to TWO decimal places.

7. Number the answers correctly according to the numbering system used in this question paper.

8. You may use a non-programmable calculator.

9. Calculations must include:

   9.1 Formulae and manipulations where needed

   9.2 Correct replacement of values

   9.3 Correct answer and relevant units where applicable

10. A formula sheet is attached at the end of this question paper.

11. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question numbers (1.1 to 1.15) in the ANSWER BOOK, e.g. 1.16 D.

1.1 An employee with a well-developed sense of teamwork …

A works well independently without any distractions.
B encourages open communication and collaboration with others.
C enjoys having a strict leader who makes all the decisions. (1)

1.2 A monostable multivibrator can … in electronic circuits.

A be used as a free running oscillator
B vary the time in both its stable states
C eliminate switch bounce
D amplify a small audio input signal to a large output signal (1)

1.3 The circuit used in an audio mixer to individually amplify or attenuate each input signal is the ...

A comparator.
B Schmitt trigger.
C inverting op amp.
D summing amplifier. (1)

1.4 A basic op-amp comparator has ... between input and output.

A no feedback
B a positive feedback loop
C a negative feedback loop
D None of the above-mentioned (1)

1.5 When the input receives a positive-going voltage waveform, the current flowing into the inverting terminal of the op-amp differentiator circuit is …

A equal to the rate of change of the input current.
B inversely proportional to the input current.
C directly proportional to the input current.
D zero. (1)

1.6 The concept of feedback in an op-amp circuit is to sample the … and feed it back to one of the input terminals.

A output voltage
B output impedance
C output frequency
D B and C (1)

1.7 The pin that activates the 555 IC is known as the … pin.

A threshold
B trigger
C discharge
D control voltage (1)

1.8 A half-adder is a combinational logic circuit consisting of a/an … gate.

A NAND gate and an EX-OR
B NAND gate and an EX-NOR
C AND gate and an EX-OR
D AND gate and an EX-NOR (1)

1.9 An active high RS latch is in the set state when …

A R=1 and S=0.
B R=0 and S=1.
C R=1 and S=1.
D R=0 and S=0. (1)

1.10 A frequency divider connects a number of flip-flops together in series. Having an initial clock frequency introduced to the first stage will produce a … from the first flip-flop.

A divide by two
B divide by one
C divide by half
D divide by four (1)

1.11 A register that allows four bits of data to be loaded simultaneously and then have data shifted out one bit at a time, is known as a … register.

A parallel-in: parallel-out
B series-in: series-out
C parallel-in: series-out
D series-in: parallel-out (1)

1.12 The unit that is responsible for carrying out calculations on the data in a microcontroller is known as the … unit.

A memory
B control
C input-output
D arithmetic logic (1)

1.13 The term RAM means … with reference to microcontrollers.

A random allowable memory
B read allowable memory
C read access memory
D random access memory (1)

1.14 A/an … bus is used to transmit the memory address from memory and input/output ports to the CPU.

A system
B data
C control
D None of the above-mentioned (1)

1.15 The basic operational process of a computer is called a/an …

A algorithm.
B flow chart.
C program.
D instruction cycle. (1)
[15]

QUESTION 2: OCCUPATIONAL HEALTH AND SAFETY

2.1 State whether it is an *unsafe act* or a *condition* when a person neglects to wear safety goggles while working with a pedestal drill. (1)

2.2 Define the term *employer* with reference to the Occupational Health and Safety Act, 1993 (Act 85 of 1993). (2)

2.3 Explain *quantitative risk analysis*. (2)

2.4 Explain the purpose of human rights in the workplace. (2)

2.5 An employee who has a sense of responsibility displays good work ethics. State how it can help the company achieve the production goals. (3)
[10]

QUESTION 3: SWITCHING CIRCUITS

3.1 Explain the term *dual voltage supply* with reference to operational amplifiers. (2)

3.2 FIGURE 3.2 below shows a bistable multivibrator using a 555 IC. Answer the questions that follow.

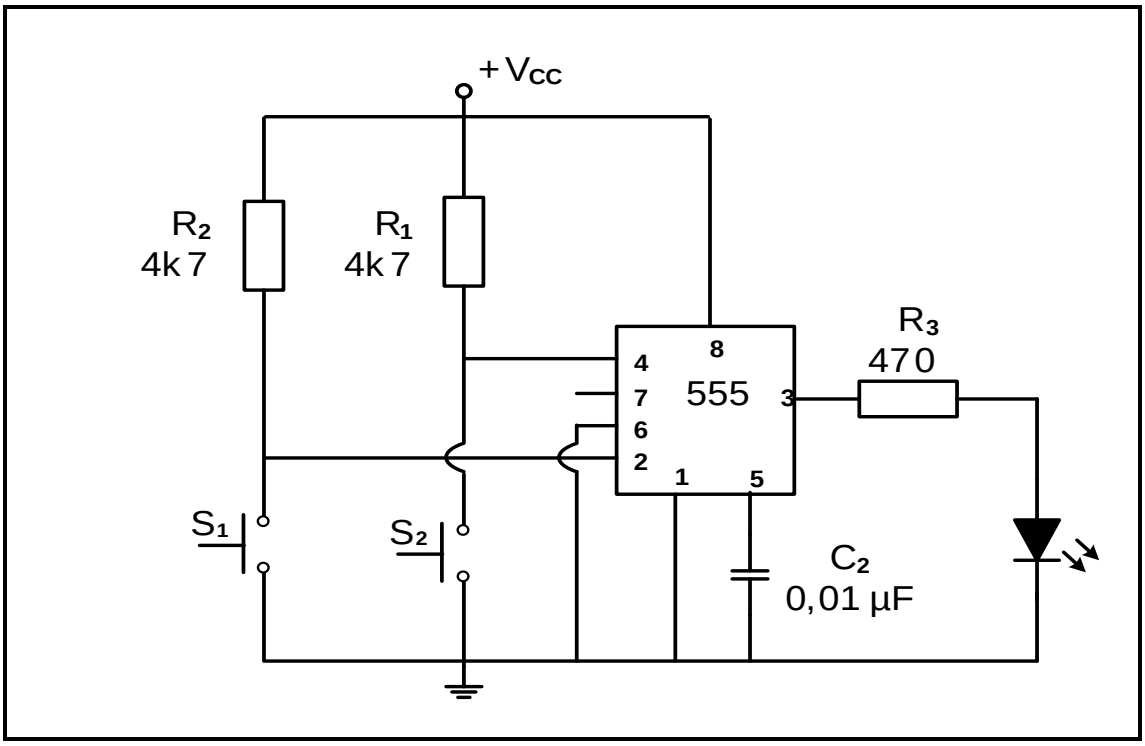


FIGURE 3.2: BISTABLE MULTIVIBRATOR

3.2.1 State the purpose of capacitor C_2. (1)

3.2.2 Explain the operation of the circuit in FIGURE 3.2 above when:

(a) S_1 is pressed (3)

(b) S_2 is pressed while S_1 remains closed (2)

3.2.3 Explain why the output of the circuit remains in a stable state once any of the switches are pressed. (2)

3.3 Refer to FIGURE 3.3 below and answer the questions that follow.

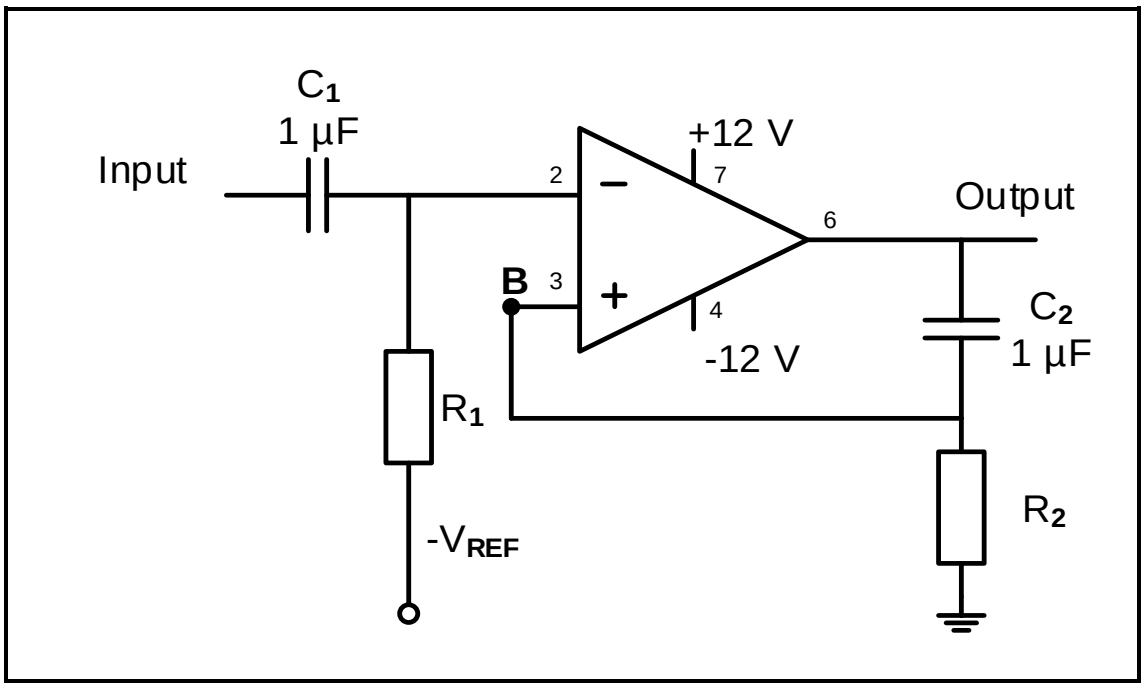


FIGURE 3.3: OP-AMP MONOSTABLE MULTIVIBRATOR CIRCUIT

3.3.1 State the purpose of $-V_{REF}$ in the inverting input during its natural resting condition. (2)

3.3.2 State what the voltage at point **B** will be when capacitor C_2 is fully charged. Motivate your answer. (2)

3.3.3 Refer to FIGURE 3.3.3 below and draw the output on the ANSWER SHEET for QUESTION 3.3.3.

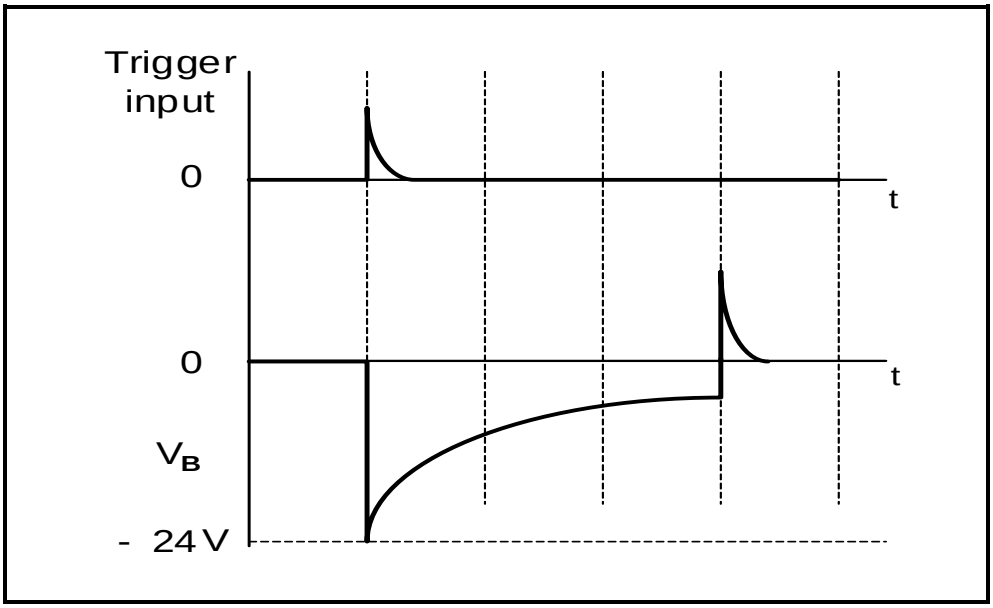


FIGURE 3.3.3: INPUT SIGNALS (4)

3.4 Explain the difference in operation between a 555 *monostable multivibrator* and a 555 *astable multivibrator*. (4)

3.5 FIGURE 3.5 below shows the input and output of a Schmitt trigger. Answer the questions that follow.

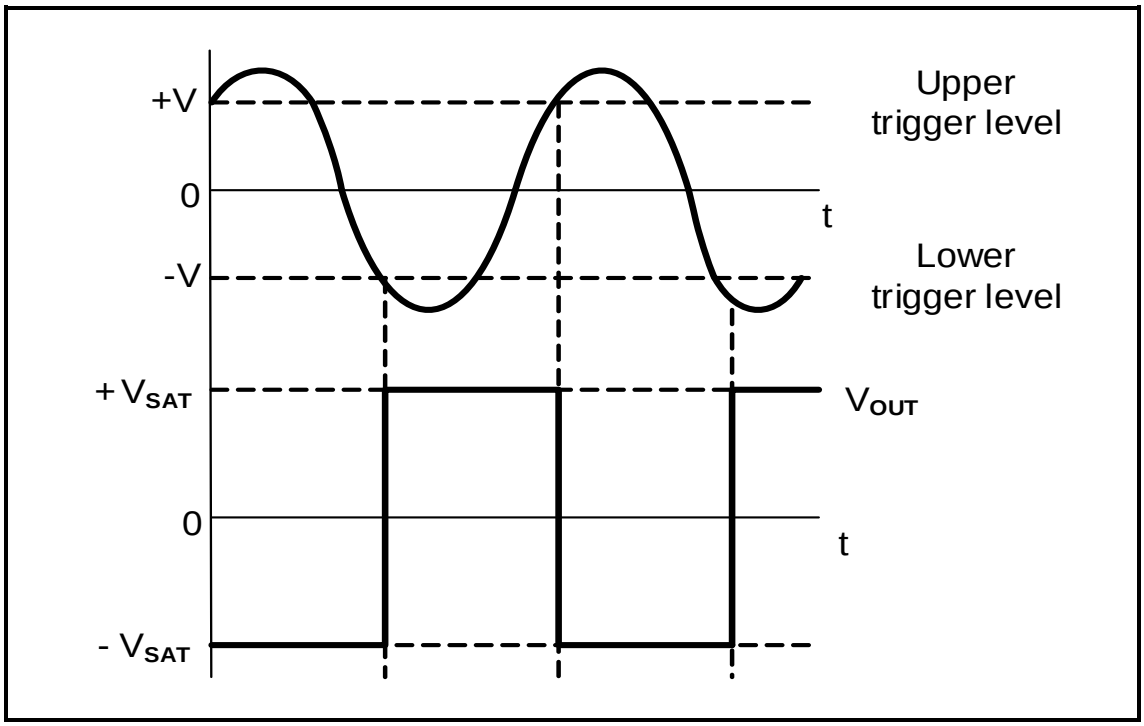


FIGURE 3.5: SCHMITT TRIGGER INPUT AND OUTPUT

3.5.1 State whether the signals provided in FIGURE 3.5 above represent an inverting or non-inverting Schmitt trigger. Motivate your answer. (3)

3.5.2 Complete the circuit diagram of a Schmitt trigger in FIGURE 3.5.2 below on the ANSWER SHEET for QUESTION 3.5.2.

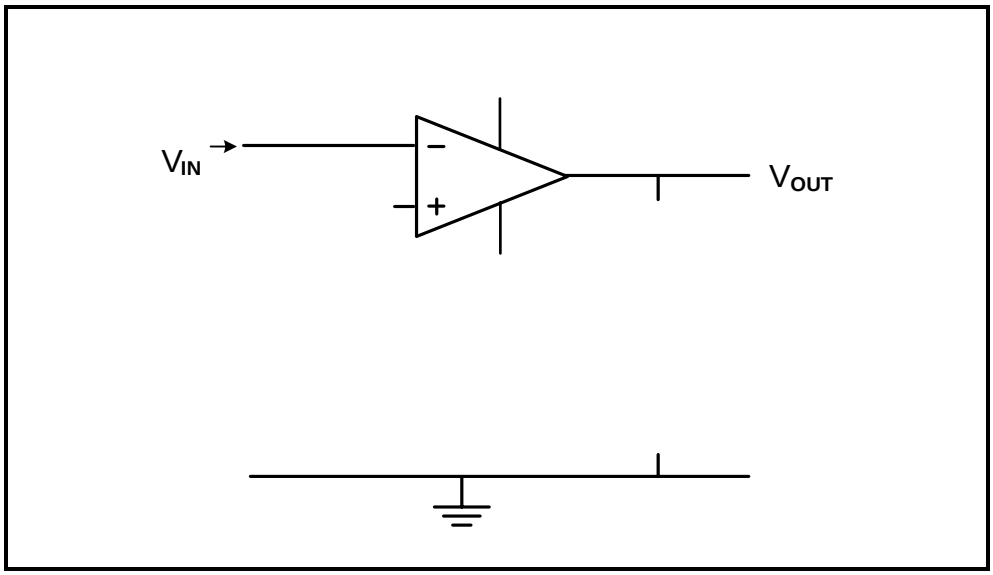


FIGURE 3.5.2: INCOMPLETE SCHMITT TRIGGER CIRCUIT DIAGRAM

(5)

3.6 Refer to FIGURE 3.6 below and answer the questions that follow.

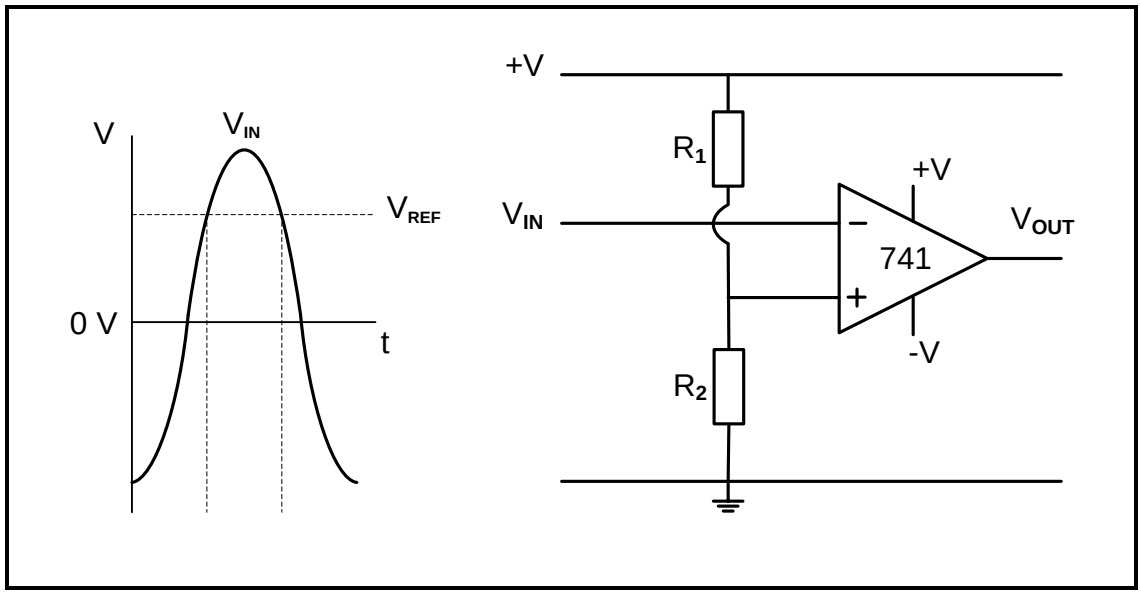


FIGURE 3.6: OP-AMP COMPARATOR

3.6.1 Explain the function of R_1 and R_2 with reference to the input voltage. (2)

3.6.2 Draw the output for the circuit in FIGURE 3.6 above on the ANSWER SHEET for QUESTION 3.6.2. (3)

3.7 Refer to FIGURE 3.7 below where V_2 is 180º out of phase with V_1 and V_3. Answer the questions that follow.

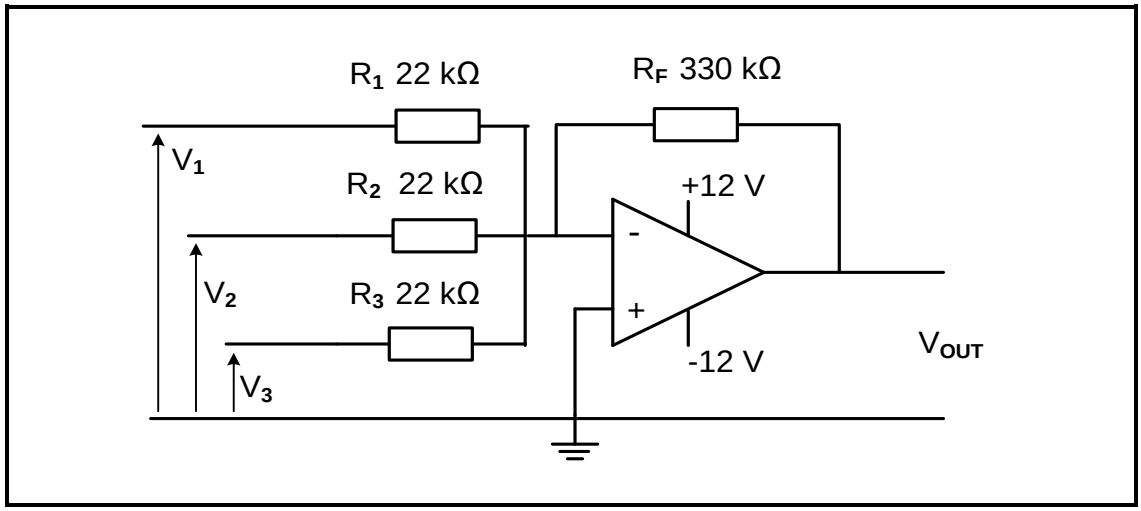


FIGURE 3.7: SUMMING AMPLIFIER

Given:

R_1, R_2, R_3 = 22 kΩ
R_F = 330 kΩ
V_1 = 0,1 V
V_2 = -0,2 V
V_3 = 0,3 V

Calculate the:

3.7.1 Gain of the amplifier (3)

3.7.2 Output voltage (3)

3.7.3 Output voltage if V_2 is changed to be in phase with V_1 and V_3 (3)

3.8 Refer to FIGURE 3.8 below and answer the questions that follow.

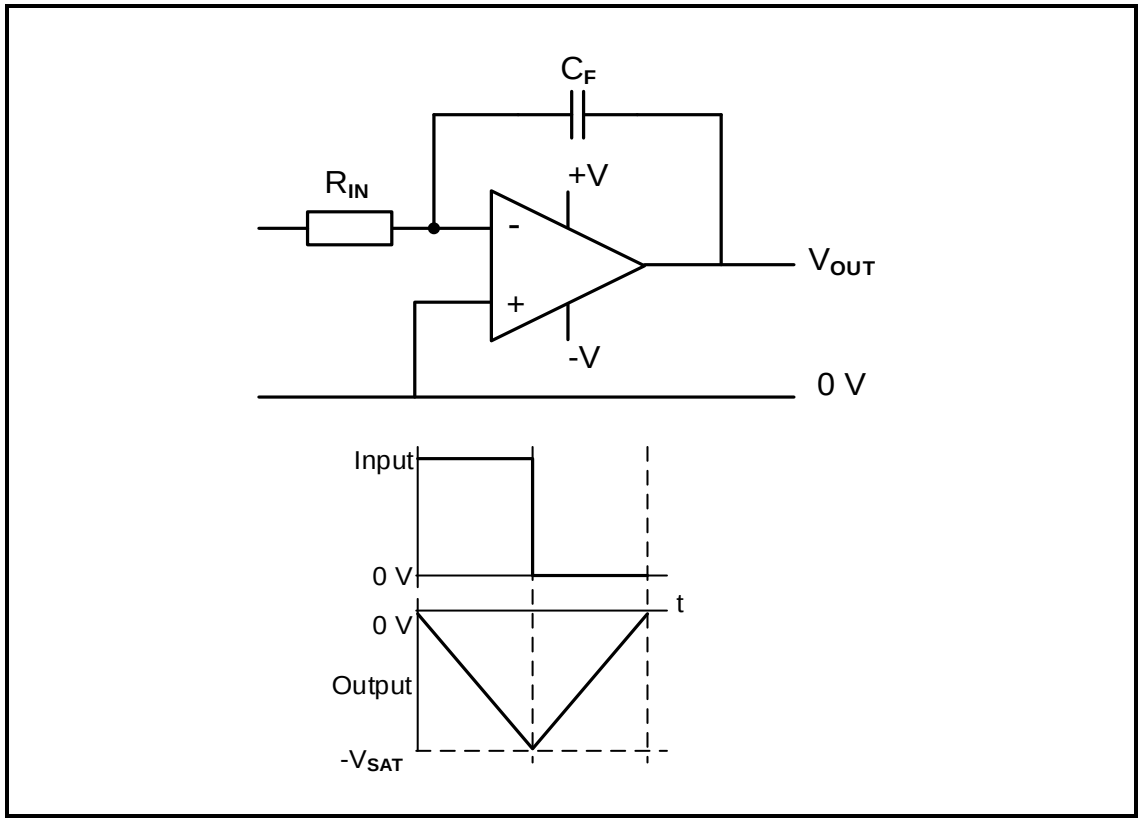


FIGURE 3.8: OP-AMP INTEGRATOR CIRCUIT

3.8.1 Name TWO key operating principles of the op-amp integrator circuit in FIGURE 3.8 above when a square wave is applied. (2)

3.8.2 Draw the output signal on the ANSWER SHEET for QUESTION 3.8.2 if the RC time constant is made shorter. (4)

[50]

QUESTION 4: SEMICONDUCTOR DEVICES

4.1 Explain *common-mode rejection ratio* with reference to a 741 op amp. (2)

4.2 Refer to FIGURE 4.2 below and answer the questions that follow.

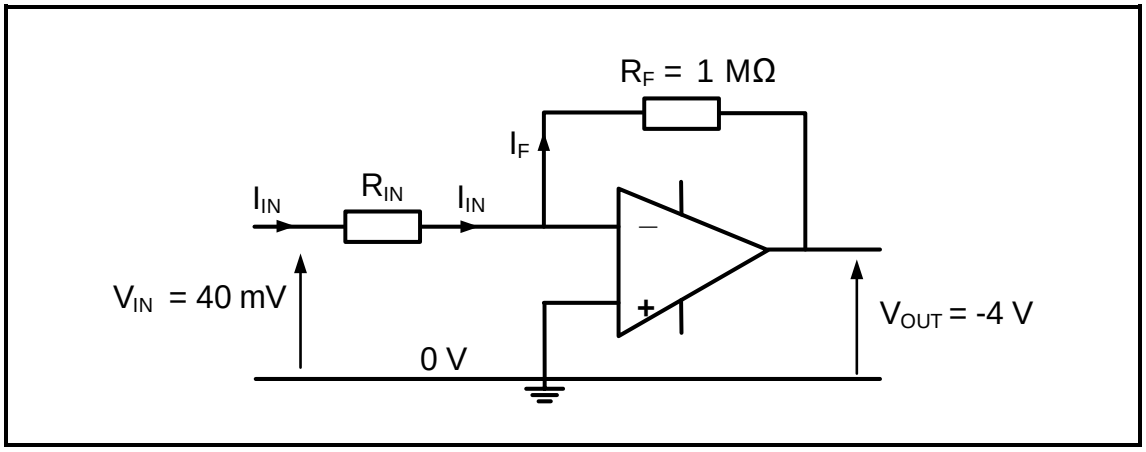


FIGURE 4.2: INVERTING OPERATIONAL AMPLIFIER

Given:

V_{IN} = 40 mV
V_{OUT} = -4 V
R_F = 1 MΩ

4.2.1 Calculate the input resistor (R_{IN}). (3)

4.2.2 State why there is no current flow into the inverting input terminal of the op amp in FIGURE 4.2 above. (1)

4.3 FIGURE 4.3 below shows a non-inverting op amp. Answer the questions that follow.

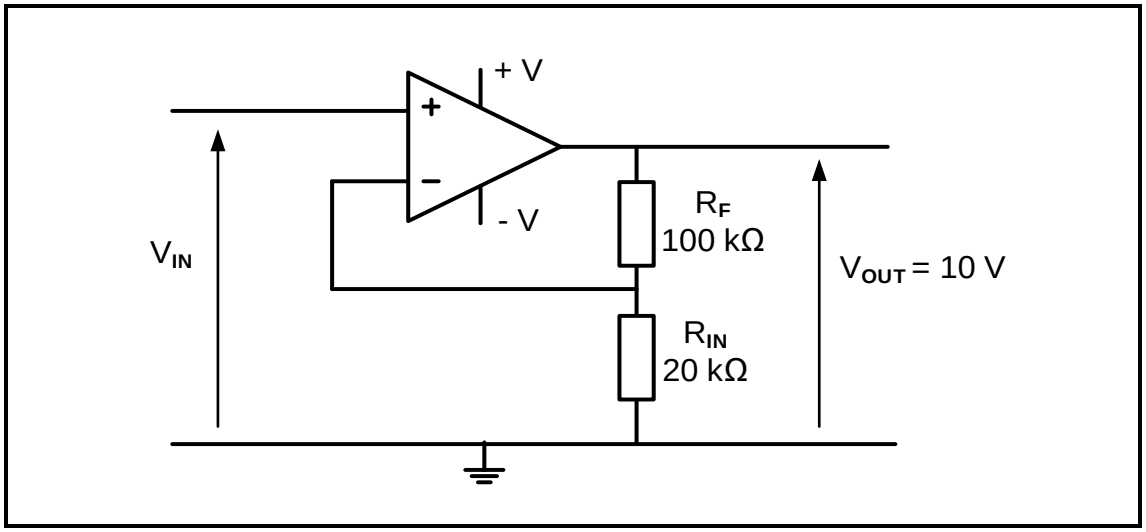


FIGURE 4.3: NON-INVERTING OP AMP

Given:

V_{OUT} = 10 V
R_F = 100 kΩ
R_{IN} = 20 kΩ

4.3.1 Name the component that can be connected between the non-inverting input terminal and the ground in FIGURE 4.3 to maintain DC stability on the op amp. (1)

4.3.2 Calculate the input voltage. (3)

4.4 FIGURE 4.4 below shows the construction layout of a 555 IC. Answer the questions that follow.

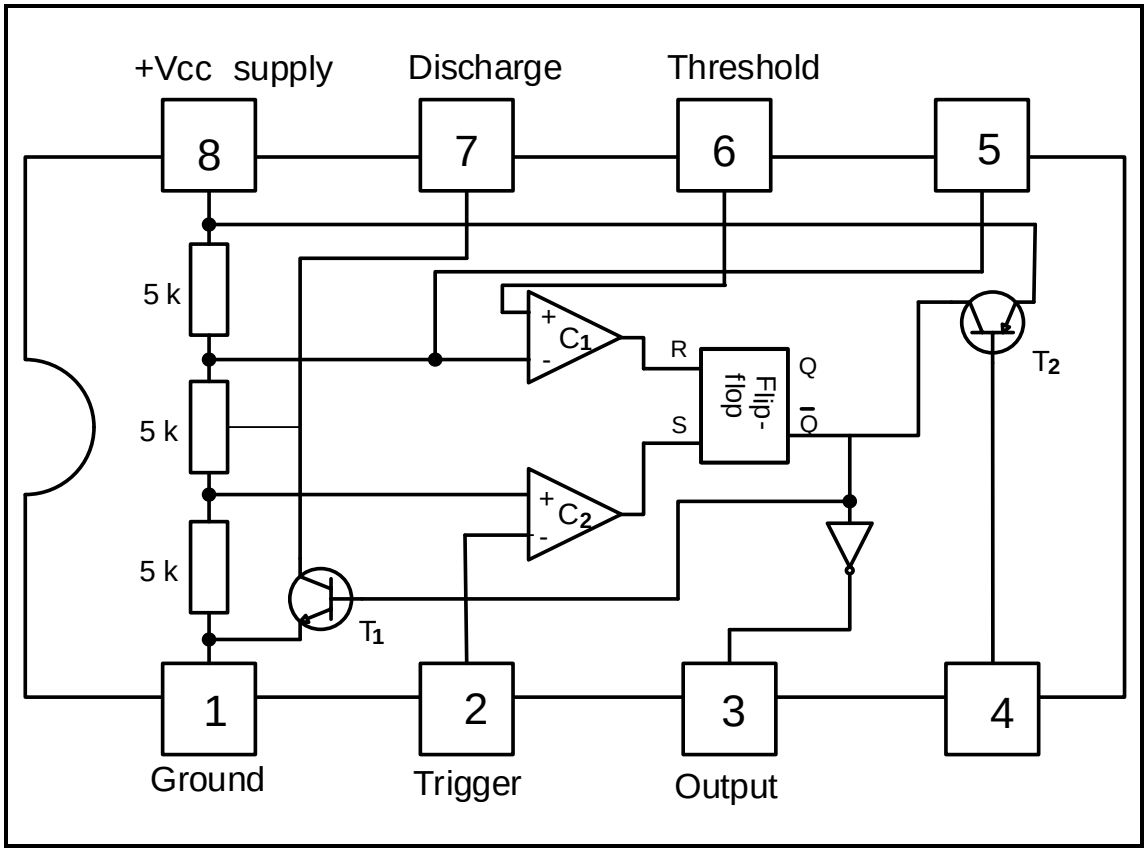


FIGURE 4.4: 555 IC

4.4.1 Label pins **4** and **5**. (2)

4.4.2 Name TWO primary building elements (blocks) of the 555 IC in FIGURE 4.4 above. (2)

4.4.3 State ONE basic application of the 555 IC other than the industrial applications, like temperature measurement and the positioning of servo devices. (1)

4.4.4 Explain the function of transistor T_1 in FIGURE 4.4 above. (2)

4.4.5 Determine the state of the outputs of the following components in FIGURE 4.4 above when the threshold voltage is greater than 2/3 V_{cc} and the trigger voltage is greater than 1/3 V_{cc}:

(a) The comparator C_1 (1)

(b) The comparator C_2 (1)

(c) $\overline{Q}$ of the RS flip-flop (1)

[20]

QUESTION 5: DIGITAL AND SEQUENTIAL DEVICES

5.1 State TWO common connections for illuminating the LEDs in the seven-segment display. (2)

5.2 Refer to FIGURE 5.2 below and answer the questions that follow.

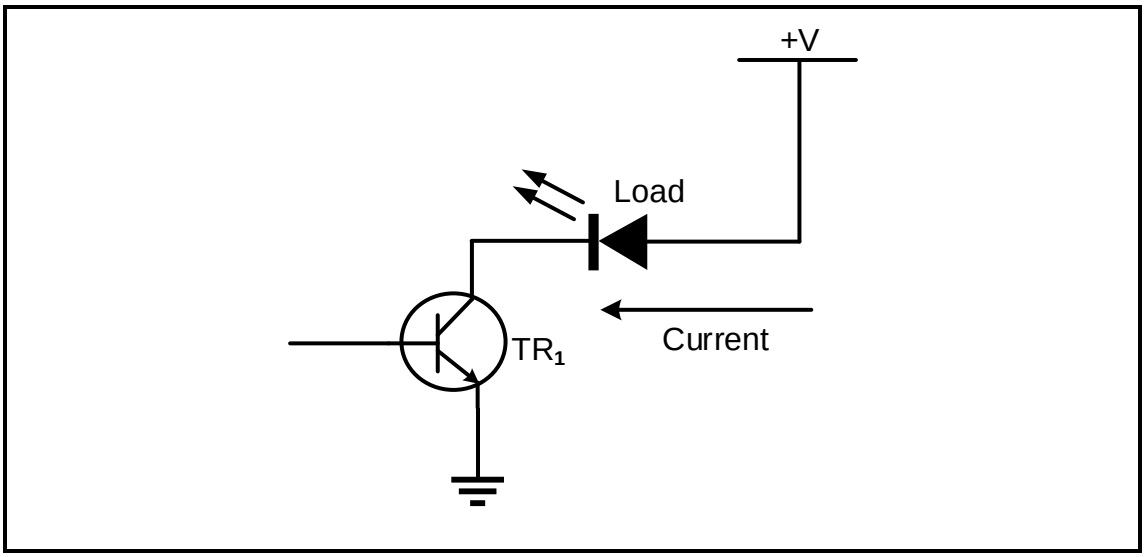


FIGURE 5.2

5.2.1 Identify FIGURE 5.2 above. (1)

5.2.2 State whether there will be current flow in the circuit when the transistor (TR_1) is replaced with a PNP transistor and an emitter is connected to the ground. Motivate your answer. (2)

5.3 Refer to FIGURE 5.3 below and answer the questions that follow.

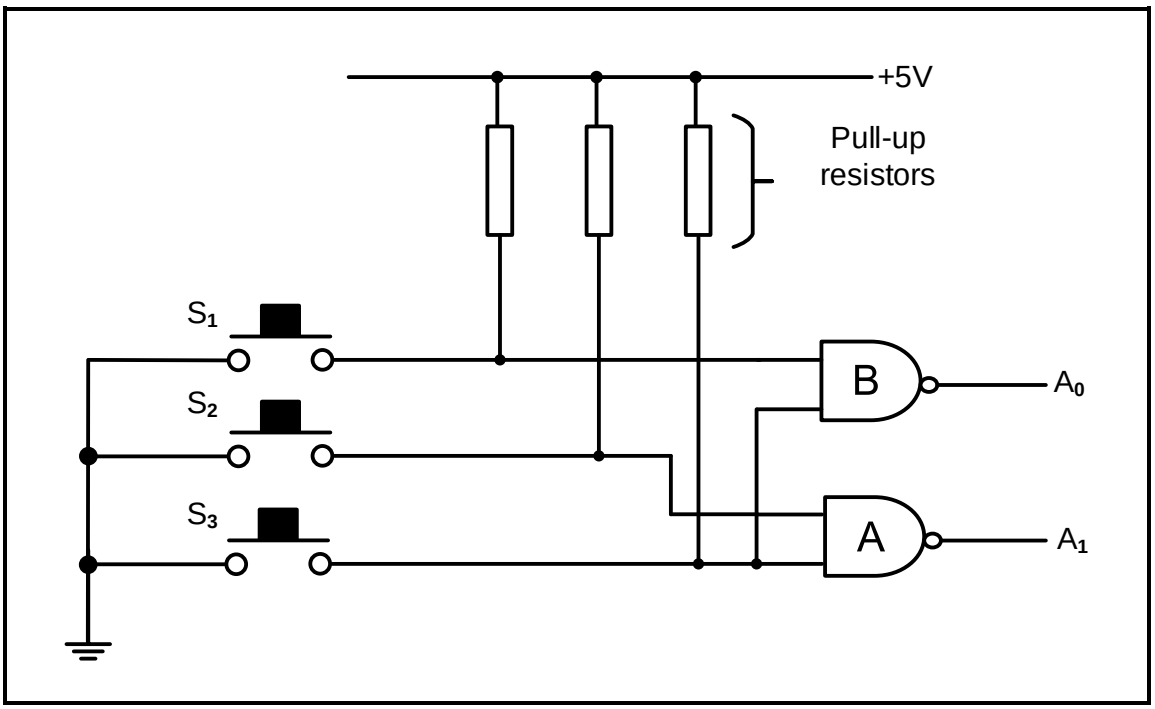


FIGURE 5.3: LOGIC CIRCUIT

5.3.1 Identify the logic circuit in FIGURE 5.3 above. (1)

5.3.2 State the function of the logic circuit in FIGURE 5.3 above. (1)

5.3.3 State the purpose of the pull-up resistors when switches S_1–S_3 in FIGURE 5.3 above are open. (1)

5.3.4 Determine the logic output in A_0 and A_1 when switch S_2 in FIGURE 5.3 above is pressed. (2)

5.4 Refer to adders as arithmetic circuits and answer the questions that follow.

5.4.1 FIGURE 5.4.1 below shows a two-bit parallel adder. Determine the output when the binary numbers 10_2 and 11_2 are added. Write the answers in squares A, B and C provided on the ANSWER SHEET for QUESTION 5.4.1.

NOTE: $A_1\ A_0 = 10_2$ and $B_1\ B_0 = 11_2$

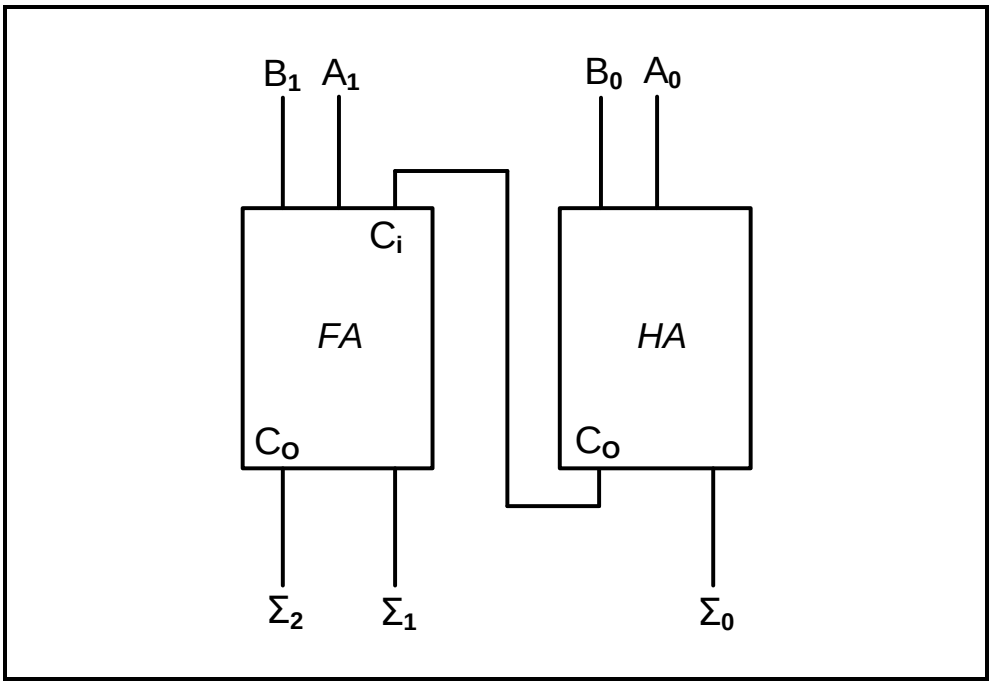


FIGURE 5.4.1: TWO-BIT PARALLEL ADDER (3)

5.4.2 Complete the drawing of a full adder logic circuit on the ANSWER SHEET for QUESTION 5.4.2. (5)

5.5 Refer to FIGURE 5.5 below and answer the questions that follow.

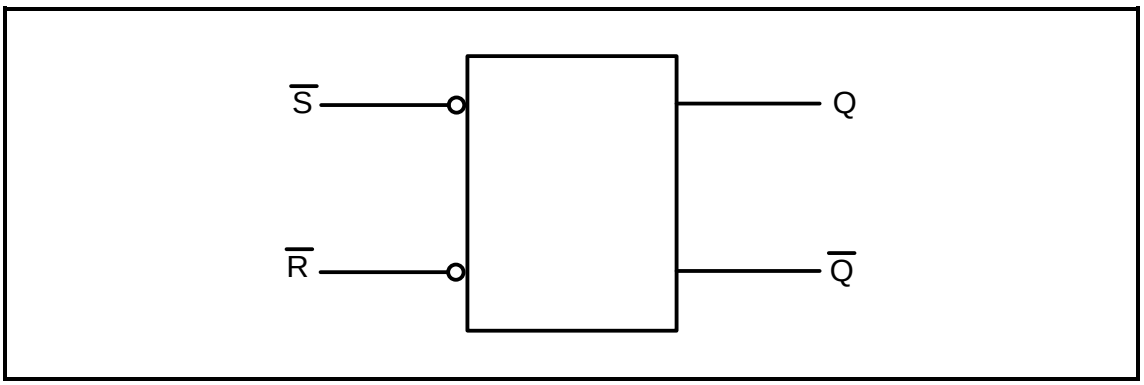


FIGURE 5.5: LOGIC SYMBOL

5.5.1 Identify the logic symbol in FIGURE 5.5 above. (1)

5.5.2 Complete the logic circuit diagram of the latch represented by FIGURE 5.5 above on the ANSWER SHEET for QUESTION 5.5.2. (3)

5.5.3 Complete the truth table of the logic circuit in FIGURE 5.5 above on the ANSWER SHEET for QUESTION 5.5.3.

$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$
0	0	Invalid	Invalid
0	1		
1	0		
1	1	No change	No change

TABLE 5.5.3 (2)

5.6 Explain the difference between *combinational logic circuits* and *sequential logic circuits* with regard to their basic building elements. (2)

5.7 Complete the logic circuit diagram of a D flip-flop represented by FIGURE 5.7 below on the ANSWER SHEET for QUESTION 5.7.

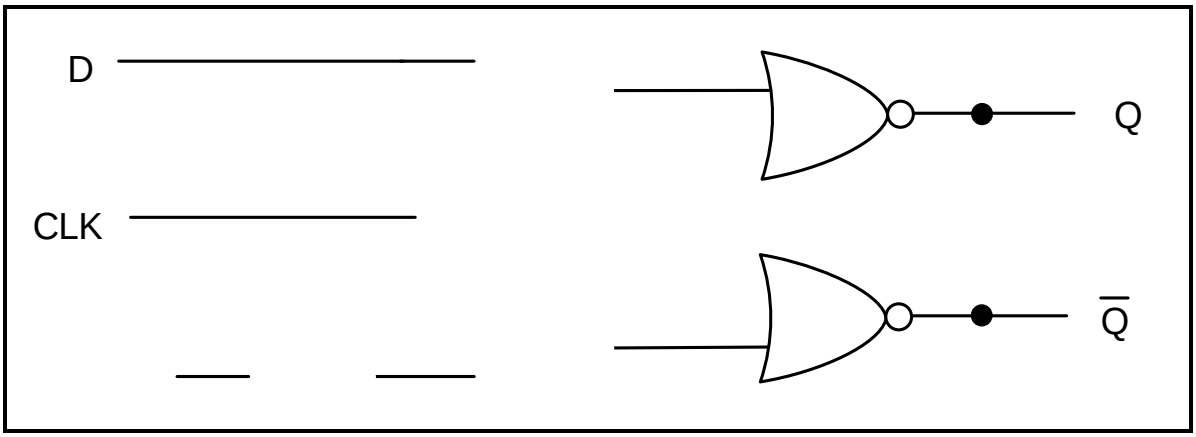


FIGURE 5.7: INCOMPLETE LOGIC CIRCUIT (6)

5.8 Refer to FIGURE 5.8 below and answer the questions that follow.

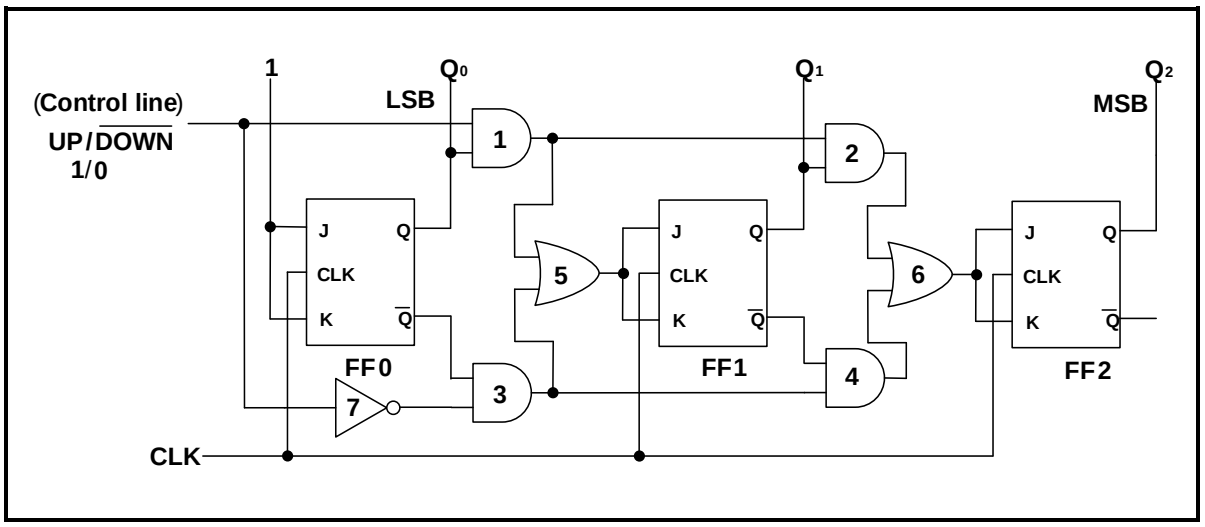


FIGURE 5.8: UP/DOWN COUNTER

5.8.1 State the output of the logic gates below when the control line in FIGURE 5.8 above is set to 'low':

(a) Gate 1 (1)

(b) Gate 3 (1)

5.8.2 Describe the function of logic gate 7 in FIGURE 5.8 above. (1)

5.8.3 Name the mode of operation when the control line is set to 'high'. (1)

5.8.4 In FIGURE 5.8 above, the output of FF0 'Q' is logic 1.

Explain the operation of the counter when the input of JK flip-flop (FF**0**) is changed to logic '0' and the control line is made logic '0'. (4)

5.9 Refer to counters and answer the questions that follow.

5.9.1 Explain *negative-edge triggered flip-flops*. (1)

5.9.2 State TWO applications of the up/down counter. (2)

5.10 Complete the logic circuit diagram of a modulus five-counter represented by FIGURE 5.10 below on the ANSWER SHEET for QUESTION 5.10.

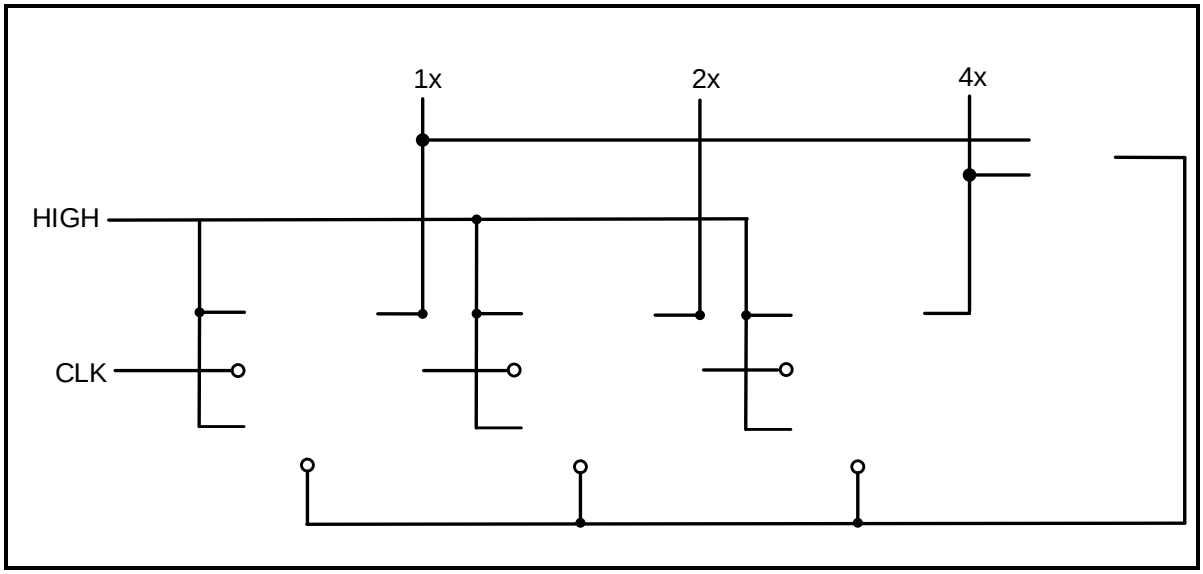


FIGURE 5.10: INCOMPLETE MODULUS FIVE-COUNTER (5)

5.11 Explain the term *propagation delay*. (2)

5.12 Refer to FIGURE 5.12 below and answer the questions that follow.

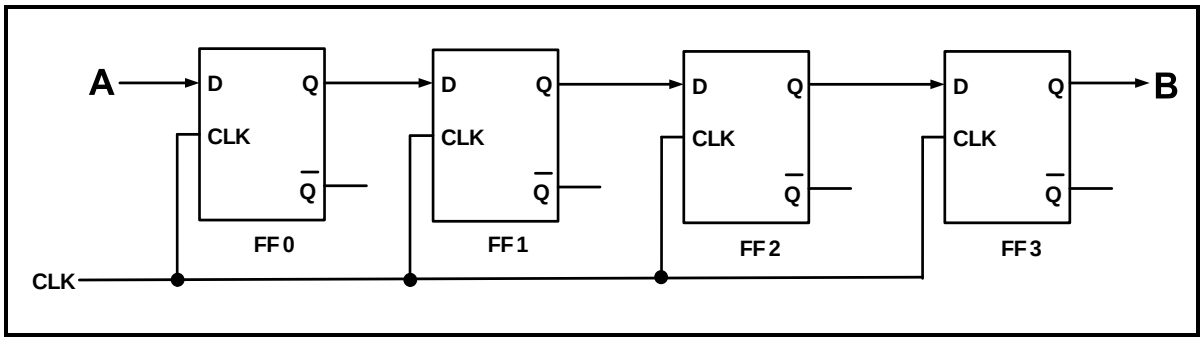


FIGURE 5.12: REGISTER

5.12.1 Identify the register in FIGURE 5.12 above. (1)

5.12.2 State how many clock pulses are needed to load and move out a four-bit number such as 1011_2. (1)

5.12.3 Describe how data flows through the register. (3)

[55]

QUESTION 6: MICROCONTROLLERS

6.1 Explain how microprocessors provide digital processing of data to devices. (5)

6.2 Refer to the block diagram of a CPU with registers in FIGURE 6.2 below and answer the questions that follow.

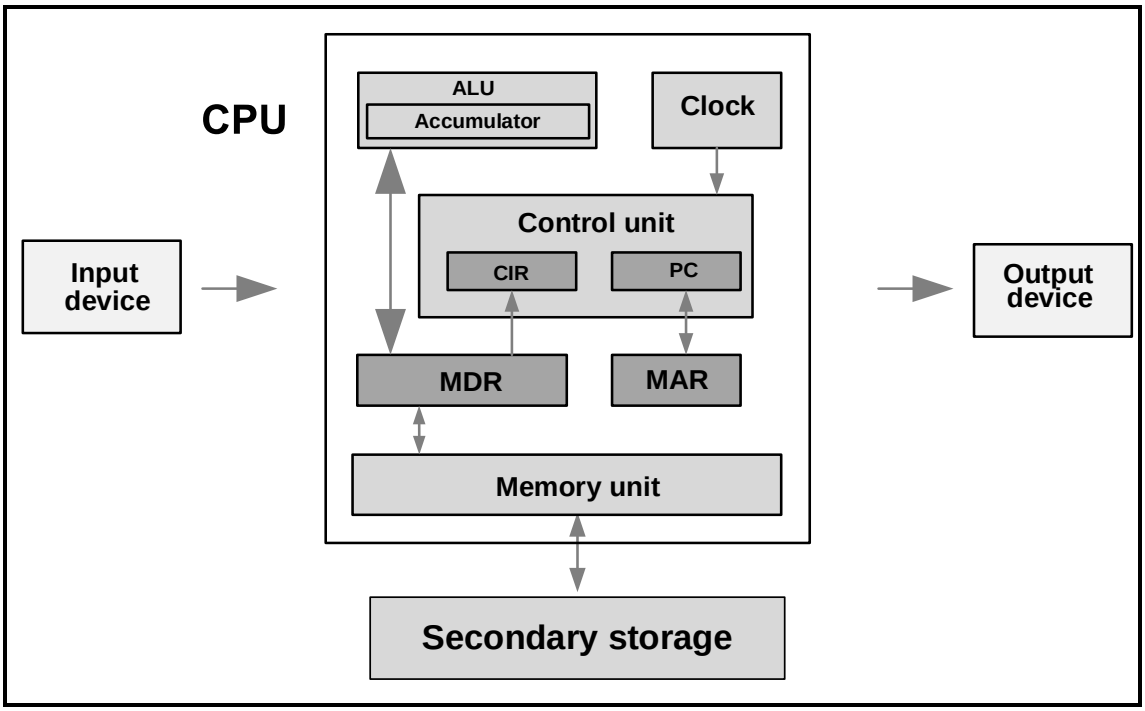


FIGURE 6.2: CPU WITH REGISTERS

6.2.1 Identify the components in FIGURE 6.2 above that are described by the following statements:

(a) The component that stores data needed as part of an arithmetic operation (1)

(b) The component that fetches each instruction to be executed in a program by issuing control signals to the hardware (1)

(c) The component that stores the address of the next instruction to be executed by the processor (1)

6.2.2 State the function of the following:

(a) Program counter (PC) (2)

(b) Memory data register (MDR) (1)

6.2.3 Explain the operating cycle of the CPU. (3)

6.3 Refer to FIGURE 6.3 below and answer the questions that follow.

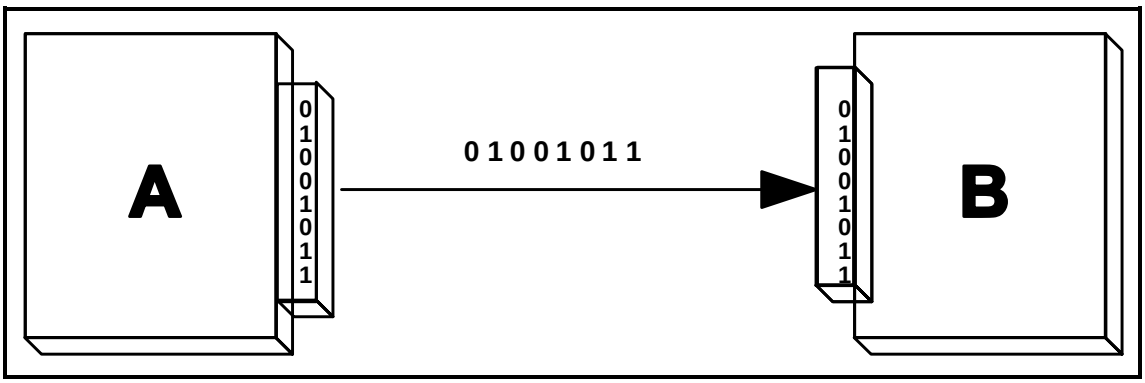


FIGURE 6.3: SERIAL DATA COMMUNICATION

6.3.1 Label **A** and **B**. (2)

6.3.2 Describe the flow of data from **A** to **B**. (1)

6.3.3 State TWO disadvantages of serial communication. (2)

6.4 Refer to communication peripherals and answer the questions that follow.

6.4.1 Name TWO advantages of the UART. (2)

6.4.2 Explain the operation of the UART. (6)

6.4.3 Name ONE communication peripheral other than the universal asynchronous receiver transmitter (UART) or the SPI. (1)

6.4.4 Describe the serial peripheral interface bus (SPI) as a type of communication peripheral. (3)

6.5 Refer to communication protocol and answer the questions that follow.

6.5.1 Define *communication protocol*. (2)

6.5.2 State TWO types of communication protocols. (2)

6.5.3 Name TWO types of serial interface connections used in asynchronous communication. (2)

6.6 State the function of an analogue-to-digital-converter (ADC) with reference to the hardware of microcontrollers. (1)

6.7 Refer to the software of microcontrollers and define the following terms:

6.7.1 Program (2)

6.7.2 Flow diagram (2)

6.8 FIGURE 6.8 on the ANSWER SHEET for QUESTION 6.8 shows an incomplete flow chart of a bi-stable device that has two stable states.

Study the following conditions and complete the flow chart on the ANSWER SHEET for QUESTION 6.8 OR alternatively, if you choose not to use the incomplete flow chart, then provide the answer in the ANSWER BOOK.

Conditions:

- The device is set by input (A) going high where it will remain until it is reset. This produced a HIGH OUTPUT.
- It is reset by input (A) going low where it will activate INPUT (B) into a HIGH state.
- A HIGH state (B) will cause the output to toggle to a low output.

The device has a single output. (8)
[50]

TOTAL: 200

Copyright reserved

FORMULA SHEET

SEMICONDUCTOR DEVICES

Gain $A_V = \frac{V_{OUT}}{V_{IN}} = -\left(\frac{R_F}{R_{IN}}\right)$ **OR** $A_V = 1 + \frac{R_F}{R_{IN}}$

$V_{OUT} = V_{IN} \times \left(-\frac{R_F}{R_{IN}}\right)$

$V_{OUT} = V_{IN} \times \left(1 + \frac{R_F}{R_{IN}}\right)$

SWITCHING CIRCUITS

$V_{OUT} = -\left(V_1 \frac{R_F}{R_1} + V_2 \frac{R_F}{R_2} + ... V_N \frac{R_F}{R_N}\right)$

Gain $A_V = \frac{V_{OUT}}{V_{IN}} = \frac{V_{OUT}}{(V_1 + V_2 + ... V_N)}$

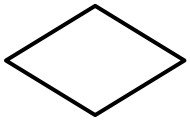
$V_{OUT} = -(V_1 + V_2 + ... V_N)$

$F = \frac{1}{T}$

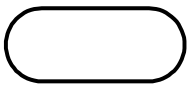
FLOW CHART SYMBOLS IN PICAXE



Process



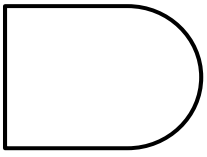
Decision



Terminator



Data



Wait

Copyright reserved